MILITARY SPECIFICATION

ENVELOPE, RADIOGRAPHIC FILM MOUNT, DENTAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers dental film mount radiographic envelopes.

1.2 Classification. The envelopes shall be of the following sizes, as specified (see 6.2).

Size 1 - 5 1/2 by 9 1/4 inches.
Size 2 - 3 1/8 by 5 1/2 inches.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code DPSC-ATT, 2800 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-L-10547 - Liner, Case, and Sheet Overwrap, Water Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

FED-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
FED-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI) STANDARDS AND SUGGESTED METHODS

T-401 Fiber Analysis of Paper and Paperboard.
T-402 Standard Conditioning and Testing Atmospheres for Paper, Board, Pulp Handsheets, and Related Products.
T-403 Bursting Strength of Paper.
T-410 Weight Per Unit Area (Basis Weight or Substance) of Paper and Paperboard.
T-423 Folding Endurance of Paper (Schopper Tester).
T-435 Hydrogen ION Concentration (pH) of Paper Extracts-Hot Extraction Method.

(Application for copies should be addressed to Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30341.)

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3. REQUIREMENTS

3.1 Material. The envelopes shall be fabricated of smooth, light-colored, heavyweight No. 1 kraft paper, composed of 100 percent unbleached sulfate pulp. No ground wood pulp shall be present in the finished paper. The fiber content shall be determined by means of the method given in 4.4.1. Recycled paper meeting the above requirements of this specification may be used.

3.1.1 Physical properties. The physical properties of the paper shall be as shown in table I for the size indicated. Physical properties shall be tested in accordance with 4.4.1.

TABLE I. Physical properties.

Size	Basis weight (17 by 22, 500 sheets) tolerance ± 5%	Folding endurance, average each direction (minimum)	Bursting strength, average (minimum)
1	28 lbs	Double folds 450	Points 60
2	20 lbs	300	36

3.1.2 Acidity (pH). The pH of the paper for the envelopes shall be not less than 5.0, when tested in accordance with 4.4.1.

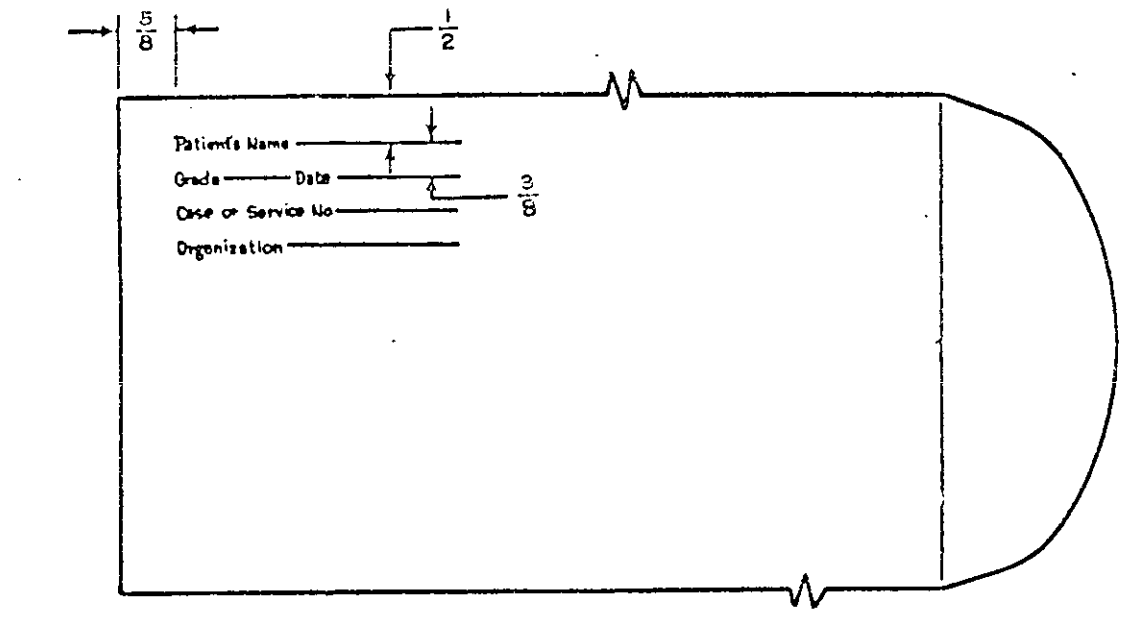
3.1.3 Accelerated aging test. Test film shall show no signs of fading, darkening or discoloration when tested in accordance with 4.4.2. The envelope shall show no signs of deterioration in accelerated aging test.

3.1.4 Stability. The decrease in folding endurance of the paper of the envelopes shall not be more than 30 percent when tested in accordance with 4.4.3.

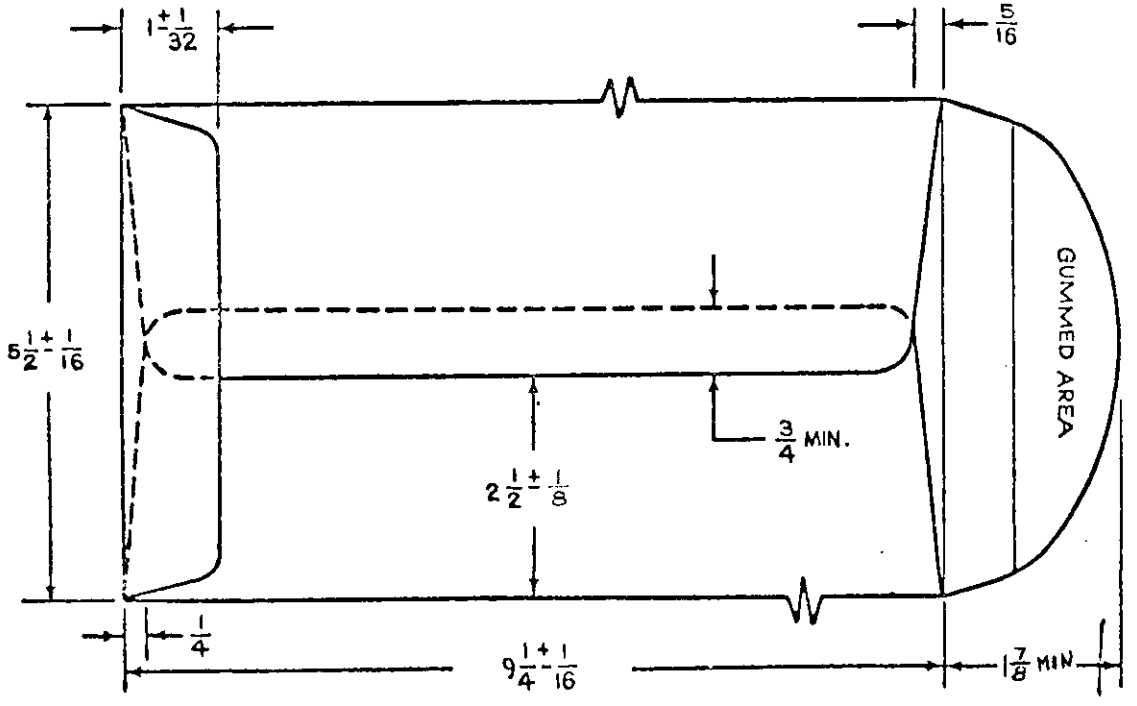
3.2 Construction. Envelopes shall be made in the center seam, open-end style. There shall be an open end along one short dimension. Style, design, and dimensions of size 1 envelope shall be as shown on figure 1. Style, design, and dimensions of size 2 envelope shall be as shown on figure 2. The insides of the flaps shall be gummed. The gum shall be of such quality that when subjected to accelerated aging test, 3.1.3, it shall not become sticky and unusable. The seams shall be securely fastened by adhesive extending approximately the whole length of each seam.

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SIZE 1



FRONT



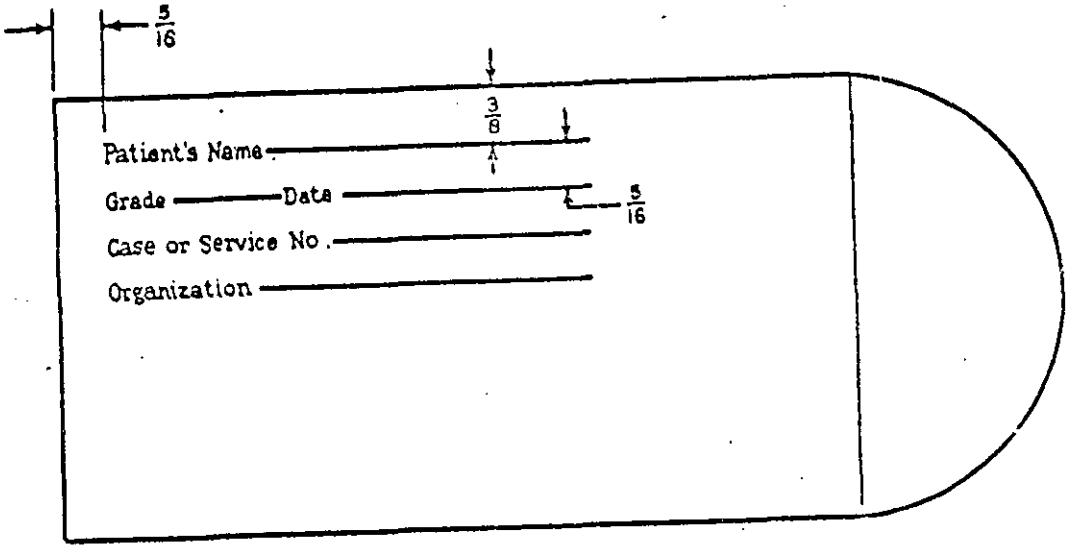
BACK

Note: All dimensions are in inches.

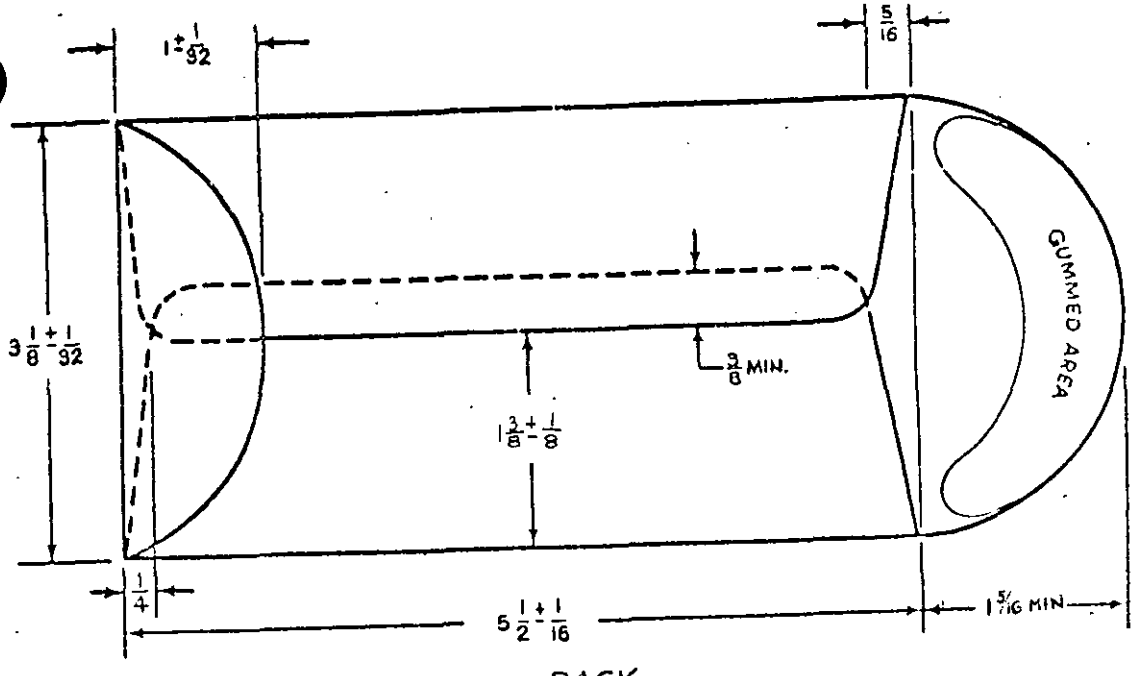
FIGURE I. Size 1 envelope.

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SIZE 2



FRONT



BACK

Note: All dimensions are in inches.

FIGURE II. Size 2 envelopes.

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3.3 Ruling and writing properties. Lines ruled and characters written shall be sharp, clear, legible and shall not feather when tested as specified in 4.5.

3.4 Printing. Printing shall be as shown on the figure for the applicable size.

3.5 Workmanship. Workmanship shall be first class throughout. The envelopes shall be free from defects which affect their appearance or which may affect their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table II. Sample unit shall be one envelope.

TABLE II. For examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	I	1.0
Minor defects	I	2.5
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III.

TABLE III. For tests.

Component (unit of product)	Characteristic (Tests)	Requirement	Test procedure	Inspection level	AQL
Envelope	Fiber identification and qualitative determination	3.1	4.4.1	S-2	*
Envelope or basic paper	Basis weight	3.1.1	4.4.1	S-2	*
Envelope	Bursting strength	3.1.1	4.4.1	S-2	*
Envelope	Folding endurance	3.1.1	4.1.1	S-2	*
Envelope	Acidity (pH)	3.1.2	4.4.1	S-2	*
Envelope	Accelerated aging test	3.1.3	4.4.2	S-2	*
Envelope	Stability	3.1.4	4.4.3	S-2	*
Envelope	Ruling and writing properties	3.3	4.5	S-2	*

* Acceptance number shall be zero.

4.3 Examination. The envelopes shall be examined to determine compliance with specification requirements.

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4.3.1 Classification of defects. Examination shall be conducted in accordance with the following table:

TABLE IV. Classification of defects.

Categories	Defects *
Major	
101	Envelope not free of tears or holes.
102	Envelope seams not adequately sealed.
103	Envelope not free of stain.
104	Envelope not free of wrinkles.
105	Printing not complete or legible.
106	Adhesive on inside of envelope.
Minor	
201	Envelope not free of foreign material.
202	Adhesive on outside of envelope.

* Examination shall not be restricted to the classified defects listed above.

4.4 Tests. Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 Physical and chemical properties. Samples shall be conditioned and tested to determine compliance with the requirements of this specification as to physical and chemical properties in accordance with the following TAPPI standards and methods.

TABLE II. Physical and chemical properties.

TESTS	TAPPI NO.
Atmospheric conditions	T-402
Basis weight	T-410
Bursting strength	T-403
Folding endurance	T-423
Acidity (pH)	T-435
Fiber identification & quantitative determination*	T-401

* Samples for the test indicated need not be conditioned.

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4.4.2 Accelerated aging test.

4.4.2.1 Physical conditions. Accelerated aging shall be achieved by incubating the test material at 50 ± 3°C and, 74 ± 2 percent relative humidity for 10 days.

4.4.2.2 Test film. The film used for test shall be processed in standard commercially available developer and fixer. The film shall be either dental radiographs or dental films exposed and processed to give a density of 0.5 to 1.5.

4.4.2.3 Procedure. The test films shall be inserted in a dental film mount of the proper size and inserted in the test envelope. The envelope shall then be bent back slightly and tied with string in order to insure contact between the inner surface of the envelope and the film and the mount. After the 10 day incubation period described in 4.4.2.1, the test film shall be visually examined.

4.4.3 Stability. Envelope shall be heated in dry air for 72 hours at 100°C and then subjected to folding endurance test of 4.4.1. Envelope shall not be conditioned for this test in accordance with T-402.

4.5 Ruling and writing properties. Using a 1-percent solution of either Color Index 18050, C.I. Acid Red, or Color Index 42085 C.I. Acid Green 3, apply ruling and written characters to the envelopes with a fine printing pen.

4.6 Packaging inspection. The inspection of the unit package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.3).

5.1.1 Unit of issue. One hundred envelopes of one size only constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit packaging. One hundred envelopes of one size only shall be clean, dry and neatly packaged in a box conformaing to PPP-B-566 or PPP-B-676. Closure of the box shall be as specified in the appendix of the applicable box specification.

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5.1.3 Level C. Envelopes shall be clean and dry. One hundred envelopes of one size only shall be packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.3).

5.2.1 Level A. Twelve unit packages containing envelopes of one size only shall be packaged in a shipping container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Twelve unit packages containing envelopes of one size only shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged envelopes shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the carrier's rules and regulations.

5.3 Marking. Interior packages and shipping containers shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Testing conditions. The testing conditions of 74 ± 2 percent relative humidity at 50 ± 3°C can readily be met by placing the specimen in a sealed chamber containing a saturated solution of sodium chloride, with temperature maintained as specified. The chief precaution is to insure an even distribution of humidity in the chamber, which may require a fan or other device to circulate the air within the chamber.

6.2 Intended use. The envelopes covered by this specification are intended for use in filing and preserving dental X-ray films.

6.3 Ordering data. The procurement document should specify the following:

(a) Title, number and date of this specification.
(b) Size required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) National Stock Number.

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6.4 This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
7530-00-890-1015	ENVELOPE, RADIOGRPAHIC FILM MOUNT, DENTAL, 5-1/2 by 9-1/2 inches.
7530-00-890-1016	ENVELOPE, RADIOGRAPHIC FILM MOUNT, DENTAL, 3-1/8 by 5-1/2 inches.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA-DM

Project No. 7530-0188

☆U.S. GOVERNMENT PRINTING OFFICE: 1977 — 703-122/383

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-E-37816 ENVELOPE, RADIOGRAPHIC FILM MOUNT, DENTAL

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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